

MARIEHAMN, Finland

WMO: 029700

Lat: 60.125N

Lon: 19.903E

Elev: 20

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.8	7.7	-2.0	4.9	2.0	4.3	6.9	9.1	25.5	32.9	23.2	33.0	3.5	350	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	76.7	65.5	73.3	64.1	70.2	62.3	68.1	74.0	65.7	71.2	63.8	68.6	8.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	95.8	72.0	63.5	87.9	68.5	61.7	82.4	66.8	32.4	74.3	30.5	71.7	29.1	68.7	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	19.8	17.7	DB	-3.9	80.4	8.3	4.0	-9.9	83.2	-14.7	85.5	-19.3	87.8	-25.3	90.6	(4)
(5)			WB	-4.4	69.7	8.2	2.6	-10.3	71.6	-15.1	73.1	-19.7	74.6	-25.7	76.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	28.6	26.6	30.8	38.6	47.7	55.4	61.7	60.1	53.3	43.8	37.8	32.5	
	DBStd	13.56	7.73	9.50	7.72	4.88	6.37	4.97	4.86	4.88	5.09	6.10	6.32	7.70	
	HDD50	3533	663	656	595	342	122	10	0	1	28	207	366	543	
	HDD65	8005	1128	1076	1060	791	538	290	125	162	352	659	816	1008	
	CDD50	1041	0	0	0	2	49	172	364	315	126	13	0	0	
	CDD65	37	0	0	0	0	0	2	24	11	0	0	0	0	
	CDH74	223	0	0	0	0	4	17	159	43	0	0	0	0	
	CDH80	22	0	0	0	0	0	0	20	2	0	0	0	0	
Wind	WSAvg	8.0	8.9	8.0	8.0	7.8	7.6	7.7	7.3	7.4	7.6	8.0	8.5	8.9	
Precipitation	PrecAvg	20.9	1.5	1.0	1.0	1.1	1.1	1.3	2.1	2.7	2.6	2.2	2.5	1.8	
	PrecMax	30.8	3.5	3.3	2.2	2.3	2.4	3.5	5.5	4.9	6.8	5.9	4.9	3.6	
	PrecMin	14.6	0.4	0.2	0.0	0.0	0.2	0.1	0.7	0.5	0.7	0.9	0.7	0.3	
	PrecStd	3.9	0.8	0.7	0.5	0.6	0.7	0.8	1.3	1.2	1.3	1.3	1.0	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	44.3	50.1	61.0	72.0	76.4	82.6	78.7	68.4	59.0	50.8	47.7	
		MCWB	40.7	39.0	42.9	48.7	59.2	61.0	67.8	67.5	61.2	54.4	48.9	45.0	
	2%	DB	41.0	41.0	46.1	55.3	67.7	71.7	78.3	74.3	65.3	55.7	48.6	44.9	
		MCWB	39.8	37.4	40.3	46.3	56.6	60.3	66.4	65.9	59.8	52.9	47.4	42.9	
	5%	DB	39.4	38.7	42.9	51.6	64.1	68.2	75.1	71.6	63.1	54.0	47.1	43.0	
		MCWB	38.3	37.0	38.5	44.2	54.2	58.9	65.2	64.2	58.6	51.5	46.1	41.3	
	10%	DB	37.7	37.1	39.7	48.3	59.6	65.2	71.7	69.1	61.2	52.9	46.0	41.3	
		MCWB	36.5	35.9	36.9	42.4	52.2	57.2	63.6	62.5	57.5	50.5	45.0	39.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.3	40.1	43.7	50.2	62.0	65.7	71.2	71.2	62.9	55.8	49.9	45.7
			MCDB	41.8	42.3	48.7	58.8	69.3	72.6	77.4	76.6	66.3	57.5	50.4	46.8
2%		WB	39.8	38.3	41.0	47.3	58.7	62.1	68.9	67.9	60.8	53.9	47.6	43.3	
		MCDB	40.4	39.9	44.7	53.6	65.5	67.7	75.0	72.4	63.9	55.6	48.3	44.2	
5%		WB	38.3	36.9	39.0	45.0	55.4	60.1	66.7	65.3	59.4	52.5	46.2	41.4	
		MCDB	39.0	38.1	41.6	50.4	61.4	65.8	73.0	69.6	62.3	54.2	46.9	42.4	
10%		WB	36.7	35.8	37.3	42.8	52.4	58.6	64.8	63.6	58.4	51.0	44.7	40.0	
		MCDB	37.5	37.1	39.6	47.1	59.0	64.2	70.6	67.5	61.2	52.6	45.4	41.1	
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.8	11.6	13.9	15.6	14.6	14.5	14.1	12.4	10.3	7.3	7.7	
		MCDBR	7.3	9.2	15.1	20.8	23.0	20.7	20.7	18.0	14.9	11.2	7.3	7.5	
		MCWBR	6.9	7.7	10.5	13.2	13.3	11.7	10.8	10.7	10.6	9.0	7.1	7.1	
	5% WB	MCDBR	6.6	7.5	12.9	19.3	20.4	16.9	16.9	14.3	12.8	9.6	6.8	7.2	
		MCWBR	6.5	6.6	9.5	13.0	13.1	10.5	9.7	9.4	9.8	8.2	6.8	7.0	
Clear-Sky Solar Irradiance	taub	0.276	0.313	0.332	0.370	0.368	0.359	0.387	0.377	0.347	0.333	0.302	0.390		
	taud	2.186	2.258	2.320	2.337	2.421	2.489	2.432	2.447	2.516	2.418	2.229	2.661		
	Ebn at Noon	156	216	252	262	273	277	265	259	248	209	142	110		
	Edh at Noon	13	22	28	33	32	31	32	30	23	19	13	9		
All-Sky Solar Radiation	RadAvg	77	257	742	1273	1732	1924	1791	1396	864	369	104	49		
	RadStd	10	48	81	117	144	123	165	119	93	56	23	7		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-193	N/A	N/A	N/A

Nomenclature: See separate page